

Cloud Microsoft Graph API Architecture & System Flow

****Target Mailbox**:** `contact@gapanchor.com`
****Target Folder**:** `Demo enquiry!` (368+ records)
****Auth Provider**:** Microsoft Entra ID (Azure AD OAuth2)
****Graph API Endpoint**:** `https://graph.microsoft.com/v1.0/`
****Last Updated**:** Real-Time Dynamic Architecture Document

1. Executive Summary & Flow Architecture

Your GapAnchor Ops Dashboard operates 100% natively in the cloud using Microsoft Graph API for



2. Step-by-Step Data Flow: How Enquiries Are Extracted

When an enquiry email arrives or when a manual/automated sync is triggered:

Step 1: Mail Arrival in Microsoft 365 Cloud

Enquiries from your website form or direct inquiries land directly in the Demo enquiry! folder

Step 2: Cloud Authentication & Token Refresh

lib/outlook-graph-sync.ts retrieves the encrypted OAuth2 refresh token from prisma.integrationC

If the access token is near expiration, the server automatically calls:

POST <https://login.microsoftonline.com/common/oauth2/v2.0/token>

and receives a fresh 60-minute Microsoft Graph Access Token without requiring manual login.

Step 3: Microsoft Graph API Query

The app sends an authenticated HTTPS request directly to Microsoft Graph API:

GET [https://graph.microsoft.com/v1.0/me/mailFolders/{folder_id}/messages?\\$select=id,subject,from](https://graph.microsoft.com/v1.0/me/mailFolders/{folder_id}/messages?$select=id,subject,from)

Authorization: Bearer <access_token>

Step 4: Intelligent Regex & Field Extraction

The server parses the HTML/plain text email body using structured regex rules:

- Participant Name: Extracted from `Name: <val>` or email sender name.
- Email Address: Extracted from `Email: <val>` or `from.emailAddress.address`.
- Phone Number: Extracted from `Phone: <val>`.
- Training Course: Normalized into standard courses (`Manhattan WMS`, `Manhattan ProAct`)
- Country: Auto-detected via phone country code (+91 India, +1 USA, +44 UK, +971 UA)

Step 5: Database Deduplication & Upsert

Before saving, lib/outlook-graph-sync.ts checks existing records in Supabase PostgreSQL:

- Matches by Email, Participant Name, or Phone Number.
- Checks if the topic & arrival date fall within a 24-hour window.
- If match found Updates existing lead timestamp & message ID.
- If new lead Inserts new `prisma.enquiry` row with status `Open`.

Step 6: Real-Time UI Update

The Enquiry Intelligence dashboard (/enquiries) fetches the updated database records and render

3. Futuristic Capabilities Unlocked by Azure AD Integration

Now that your dashboard has direct, authorized access to Microsoft Graph API for contact@gapanc

mindmap

root((Azure AD & Graph API))

Mailbox Operations

Multi-Folder Monitoring

Automated Email Responses

Real-Time Push Webhooks

Calendar & Scheduling

Bi-Directional Outlook Sync

Automated Demo Booking

Trainee Invite Management

Team Collaboration

Microsoft Teams Notifications

Lead Assignment Webhooks

AI & Intelligence

Automated Lead Scoring

Sentiment Analysis

A. Real-Time Push Webhooks (Zero Polling Needed)

- Current: App fetches emails on-demand or via background cron.
- Next Level: Subscribe to Graph API Push Notifications (`POST /v1.0/subscriptions`).
- Benefit: The instant a new email hits `contact@gapanchor.com`, Microsoft Graph fires

B. Automated Email Replies & Auto-Quotes

- Capability: Use Graph API `POST /v1.0/me/sendMail`.
- Use Case: When a high-quality lead arrives for *Manhattan ProActive* or *Blue Yonder

C. Bi-Directional Outlook & Teams Calendar Sync

- Capability: Access `/v1.0/me/events` and `/v1.0/me/calendarView`.
- Use Case:
 - Automatically create calendar events for training sessions when assigned.
 - Generate Teams video call links directly inside the Ops Dashboard.
 - Prevent double-booking by syncing trainer schedules across Outlook & Google Calendar.

D. Multi-Folder & Mailbox Automation

- Capability: Query any folder in `contact@gapanchor.com` (`Inbox`, `Job Support`, `Inv
- Use Case: Monitor job support requests, invoice payment confirmations, or client feed

4. Bypassed Legacy Scripts Summary

Feature	Legacy Local Script	Current Azure Graph Cloud Architecture
Execution	Required local Windows Desktop Outlook & Python COM	Runs 100% on Next.js Server
Authentication	Local Windows MAPI Session	Secure OAuth2 Bearer Tokens via Azure AD
Dependency	Desktop PC must stay turned on	Server runs 24/7 in Cloud / Vercel / Supabase
Speed & Reliability	High failure risk if Outlook app closes	Instant HTTPS Graph API query
Scalability	Single local machine only	Multi-user, multi-tenant enterprise architecture